# SUSEVAL: The Effect of Administration Modality on Usability Perceptions Measured Using the System Usability Scale

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## Abstract

The System Usability Scale (SUS) is one of the most widely used instruments for measuring perceived usability, yet its standard written administration may present challenges for respondents who are more comfortable processing questions orally in English. This study examines how oral versus written administration of the "Standard" English SUS affects participants' experiences and SUS scores, using a within-subjects design in which participants evaluated six systems across two administration conditions, with quantitative analysis comparing SUS scores and internal consistency across administration modes alongside Reflexive Thematic Analysis of participants' experiences during the oral administration. Oral administration produced a mean SUS score 1.93 points lower than written administration, a difference that was not statistically significant, while both conditions demonstrated high internal consistency (Cronbach's alpha = .93 oral, .88 written). The qualitative findings indicate participants perceived interviewer presence as having both positive effects, such as allowing for clarification and justification of answers, and negative effects, including perceived pressure to respond. Given the small sample, which consisted predominantly of participants who reported fluent or native English proficiency, these findings provide preliminary evidence that oral and written administration can produce broadly comparable SUS scores while differing in how participants engage with the questionnaire, providing a basis for future research on the adaptation and localisation of usability instruments in contexts with greater linguistic and educational diversity.

## Keywords

System Usability Scale, usability evaluation, administration mode, interviewer-administered questionnaire

## 1 Introduction

The System Usability Scale (SUS) is a ten-item questionnaire that has become widely adopted for measuring the perceived usability of software and everyday products [7, 21]. Its popularity arises from its quick administration and simple computation of a usability score ranging from zero to one hundred [24]. The SUS is commonly administered using a standard written form of English, which reflects the linguistic context in which it was developed. These norms are often treated as an idealised form of English, which we refer to as "Standard" English [12]. These assumptions may become relevant when usability instruments are administered outside their original linguistic contexts.

This issue is especially relevant in South Africa, which has twelve official languages [36]. In the Western Cape, the most widely spoken languages include Afrikaans, isiXhosa, and English. Although English is widely used in formal settings such as universities and the workplace, it is not the first language of a majority of residents in the province [1]. South African English differs from "Standard" English used in many internationally developed tools [17]. For questionnaires, such as the SUS, unfamiliar wording may affect how respondents interpret items and in turn influence their scores [14]. This raises the possibility that administration modality may influence SUS responses in addition to perceived usability.

Oral administration may reduce some of the difficulties associated with written questionnaire administration by allowing respondents to process items verbally and seek clarification from an interviewer. Interviewer presence may also influence participants' comprehension, engagement, and responses. Despite the widespread use of the SUS, little is known about how administration mode affects responses in linguistically diverse contexts. Although translated and culturally adapted SUS versions exist for other languages [33, 38], it remains unclear whether oral administration of the "Standard" English SUS produces scores comparable to written administration in South African contexts.

This research addresses the following questions:

**RQ1:** How does the mode of administration (oral versus written) relate to SUS scores and internal consistency for the "Standard" English SUS?

**RQ2:** How do participants experience the presence of an interviewer during oral SUS administration in terms of comprehension, engagement, and responding, and how do these experiences help interpret the findings in RQ1?

We investigate these questions using a within-subjects design in which participants evaluate six systems across written and oral SUS administration. Quantitative analysis compares SUS scores and internal consistency across administration modes, while Reflexive Thematic Analysis [5] of the oral condition examines participants' experiences during oral administration. These provide both quantitative evidence of whether administration modality is associated with differences in SUS scores and qualitative insight into how respondents interpret and engage with the questionnaire.

The study contributes preliminary evidence on the oral administration of the "Standard" English SUS in a South African context and provides a basis for future research on adaptation and localisation of usability instruments in contexts with greater linguistic and educational diversity.

## 2 Related Work

### 2.1 The System Usability Scale

The System Usability Scale was first proposed by Brooke [7] in 1986 as a ten-item questionnaire for measuring perceived usability of

a system. It remains widely used due to its low cost and ease of administration. Respondents rate each item on a five-point Likert scale, producing an overall perceived usability score ranging from zero to one hundred. Odd-numbered items are worded positively and even-numbered items are worded negatively; this was intended to reduce response and acquiescence biases [37]. Drew et al. [13] found that the SUS scores reflect respondents' perceived success with the system rather than the system's objective usability, indicating it is an attitudinal measure of perceived usability. Kortum and Bangor [21] extended the SUS to fourteen everyday products, demonstrating it can be applied to a range of interactive products beyond software, as well as providing benchmarks for interpreting SUS scores.

## 2.2 Psychometric Properties of the SUS

Early psychometric research on the SUS suggested that it comprised separate usability and learnability dimensions [25]. Later research found that, despite statistically having a bi-dimensional structure, this was largely attributed to the alternating positive and negative wording of its items and therefore recommended treating the SUS as a unidimensional measure of perceived usability [26]. Research adapting the SUS to non-English contexts also supported this recommendation [8, 33]. Internal consistency has generally been high in these adaptations of the SUS, with Cronbach's alpha ranging from around 0.75 to above 0.90. The negatively worded items were seen to introduce difficulties in interpreting the questionnaire in the Spanish and German adaptations [8, 33].

## 2.3 SUS with Non-Native English Speakers

The SUS was originally developed in English by John Brooke in the United Kingdom [6]. This linguistic background is particularly relevant when using the SUS, as Finstad [14] identified difficulties associated with wording unfamiliar to non-native English speakers, such as "cumbersome" in SUS item 8. Nantel and Glaser [30] found that presenting an interface in a language other than a user's native language lowers their perceived usability even among users who are proficient in that language. SUS scores in linguistically diverse contexts may therefore reflect not only respondents' perceived usability of the system but also the linguistic processing required to understand the instrument.

## 2.4 Localisations of the System Usability Scale

Several linguistic adaptations of the SUS have been validated, including Spanish, German, Swedish, and Brazilian Portuguese [2, 8, 27, 33]. These studies primarily address translating the wording of the SUS rather than administration mode.

## 2.5 SUS in a Sub-Saharan African Context

Research examining the use of the SUS in Sub-Saharan Africa remains limited. Kavuma [18] considered using the SUS to evaluate Electronic Medical Record systems across Sub-Saharan Africa but instead opted for a similar five-point Likert-scale questionnaire, highlighting the limited use of the SUS within the region. In South Africa, Mujinga et al. [29] applied the SUS to evaluate an online banking biometric system, demonstrating that the instrument has been documented as being used in a South African context. To our knowledge, there has been no version of the SUS that has been translated or validated in any South African language or South African English variety.

## 2.6 English in the South African Higher Education Context

English remains dominant in formal and academic settings in South Africa, despite the linguistic diversity of the population [17]. In the higher education context, the use of English as the primary language for communication can create challenges for multilingual students [32]. Research at Stellenbosch University highlights that English occupies a dominant position in academic contexts despite the multilingual backgrounds of many students [23]. This linguistic context is relevant to questionnaires administered in English, particularly where participants have diverse linguistic backgrounds.

## 2.7 Administration Effects

Differences in questionnaire responses across administration modes are commonly referred to as mode effects. Research comparing self-administered and interviewer-administered questionnaires found differences in data quality and response patterns, including potential interviewer, acquiescence, and social desirability biases [4, 9, 16]. Oral interviewer-administered questionnaires can facilitate comprehension by allowing respondents to ask questions and receive clarification during the interview process [11]. This is relevant to the SUS, as Finstad [14] previously identified difficulties with unfamiliar wording among non-native English speakers. This introduces the possibility that the interviewer may influence how respondents engage with or interpret the SUS items.

## 2.8 Psychometric Instruments Administered in Different Modes

Research on other psychometric instruments found that administration mode does not produce consistent effects across instruments. Pérez-Salas et al. [35] found generally similar scores and internal consistency between self-administered and read-aloud conditions across four psychological instruments, although one instrument showed fewer participant disengagement behaviours in the read-aloud condition. Kendrick and Hatzenbuehler [19] found some differences across scales when administering the Minnesota Multiphasic Personality Inventory (MMPI) orally versus in writing, but considered their clinical significance to be minimal. Although this literature mainly focuses on psychological instruments rather than usability instruments, it shows that changing the administration mode can affect the measurement outcomes, while also suggesting these effects are not consistent across instruments.

## 2.9 Interviewer Effects

Interviewers are a known source of additional variance in survey estimates, commonly referred to as interviewer effects [9, 40]. Klausch et al. [20] reported higher category thresholds for attitudinal items and Likert scales when questionnaires were administered by interviewers compared to self-administered modes. Interviewer presence may help respondents remain focused on the questionnaire

and reduce behaviours such as item non-response [9]. Interviewer-administered modes have also been associated with higher social desirability bias than self-administered modes [4, 22]. These effects suggest that an interviewer may form part of the response process itself rather than simply administering the questionnaire.

### 2.10 WhatsApp as a Data Collection Method

WhatsApp has been used as a data collection medium in various African contexts [28, 39]. Researchers have used WhatsApp for qualitative data collection through features such as messaging, voice calls, and voice notes, with the platform supporting both synchronous and asynchronous interaction [10, 28, 31]. This makes WhatsApp a relevant medium for collecting oral responses, as participants do not require face-to-face interaction and can respond at times that are convenient to them [28]. The use of WhatsApp does introduce some practical considerations, including the cost of data and privacy concerns [31, 39].

## 3 Methodology

### 3.1 Study Design

This study employs a within-subjects, mixed-methods design to investigate how the mode of administration relates to responses to the "Standard" English SUS. Six systems were selected for evaluation. Each participant completed the SUS in both written and orally administered conditions, with three systems evaluated in each condition, resulting in six total evaluations per participant. A within-subjects design was selected because each participant completed both administration modes of the instrument, allowing for a direct comparison between the oral and written SUS responses. However, the repeated administration of the SUS may introduce order and learning effects, as participants may become more familiar with the SUS items. To reduce the association between a particular system and a single administration condition, different systems were evaluated in the two administration conditions. The quantitative component examined SUS scores and internal consistency across the two administration modes. The qualitative component examined participants' experiences during the oral administration, including comments made while responding to the SUS and responses to questions asked after the session. The qualitative component was used to provide insight into participants' comprehension, engagement, and interaction with the interviewer. The study was administered remotely via WhatsApp, with study communication conducted through a WhatsApp Community. This was done to allow participants to be sorted into different condition groups based on their home language(s), as this study was a part of a larger study conducted in parallel with two other studies evaluating isiXhosa and Afrikaans versions of the SUS. The written SUS was completed through Microsoft Forms and the oral SUS through WhatsApp calls. This reduced the need for in-person participation and allowed the study to be conducted remotely.

### 3.2 Participants and Recruitment

Participants were recruited from the University of Cape Town and the general public through personal connections, snowball sampling, and university recruitment channels. At UCT, invitation letters were sent via email to staff through Human Resources and to students through the Department of Student Affairs. Recruitment materials were distributed through WhatsApp, including a WhatsApp Community, where participants were contacted via direct messages to arrange participation. The inclusion criteria for this study required participants to have a minimum level of basic English proficiency. Participants self-reported their English proficiency and home language(s) through a demographic form (the options were: none, basic, conversational, fluent, and native). The recruitment materials were distributed primarily in English via the WhatsApp community. The final sample consisted of UCT students, the researchers' personal connections, and external participants from the snowball recruitment. This recruitment strategy was intended to reach participants with different linguistic backgrounds.

**Table 1: Overview of the study design**

| Component | Description |
|---|---|
| Design | Within-subjects, mixed methods |
| Participants | 25 participants |
| Administration Conditions | Written English SUS and oral English SUS |
| Systems Evaluated | 6 systems |
| Systems per Participant | 3 systems in the written condition and 3 systems in the oral condition |
| Quantitative Outcomes | SUS scores and internal consistency |
| Qualitative Data | Comments made during oral administration and responses to post-session questions |
| Written Administration | SUS items presented using Microsoft Forms |
| Oral Administration | SUS items read aloud during a WhatsApp call |

### 3.3 Materials

The primary instrument used in the study was the "Standard" English version of the System Usability Scale (SUS) (see Appendix A). The same wording was used in both administration conditions to ensure that the primary difference between conditions was the administration mode rather than the wording of the instrument. Each system evaluated by a participant was accompanied by a short task instruction to provide them with a specific interaction when completing the SUS. For physical systems such as POS machines and ATMs, participants were asked to recall a previous interaction with the system. The task instructions for each system are provided in Appendix B. Microsoft Forms was used to collect demographic information and written SUS responses. This provided a consistent format for collecting responses and allowed participants to complete the written SUS remotely. The oral SUS was administered through WhatsApp calls, with the researcher reading each SUS item aloud and recording the participants' responses in a separate Microsoft Form for the oral condition. Responses from both conditions were exported from Microsoft Forms to an Excel sheet for data analysis.

### 3.4 Systems under Evaluation

Six systems were selected for evaluation:

(1) ATMs
(2) EskomSePush
(3) Google Search
(4) Point of Sale (POS) Machines
(5) Takealot
(6) Uber

These systems were selected to provide participants with a range of digital applications and physical devices which are commonly used across South Africa. ATMs and Point of Sale (POS) machines were treated as generic categories, since participants may have interacted with different models of these devices. Google Search was selected for its widespread use. Uber, Takealot, and EskomSePush were selected to represent different types of digital applications. This selection provided variation in the types of systems evaluated while allowing participants to recall experiences with systems they had likely encountered in the recent past.

## 3.5 Ethical Considerations

The study was designed to minimise the collection of sensitive personal information. The demographic form collected basic information such as name, phone number, gender identity, province of residence, English proficiency, and home language(s). Only relevant data were retained for research purposes, while participants' phone numbers were retained only for contact purposes. Microsoft Forms was used for all form-based data collection as it is compliant with the Protection of Personal Information Act (POPIA). During data analysis, participants were assigned unique participant IDs and identifying information was kept separate from research data. The use of WhatsApp raised privacy concerns as it is a third-party software owned by Meta. No personal information was intentionally transmitted over WhatsApp beyond what was required for the study. The WhatsApp Community announcement channel feature was used to prevent members from viewing the member list and their respective phone numbers, which could only be viewed by administrators and members who already had the participant's number saved. Participants were also informed about the use of WhatsApp for study communication and the oral administration. Oral sessions were recorded only where participants provided consent. For sessions where recording consent was not provided, relevant comments and observations were documented using researcher field notes, while the post-session questions were transcribed as participants answered them. Research data was securely stored, and access was restricted to only the researchers. Ethics approval for the study was obtained from the UCT Faculty of Science Ethics Committee prior to data collection. Participants were provided with information about the study and its procedures before providing informed consent. Participation was voluntary, and participants were informed that they had the right to withdraw from the study at any time without penalty. Approval was also obtained from Human Resources (HR) and the Department of Student Affairs (DSA), allowing UCT staff and students to participate in the study.

## 3.6 Pretesting

The oral administration procedure was piloted with three participants, who each completed both the written and oral English SUS conditions and answered follow-up questions about their experience. Pretest SUS scores were excluded from the main analysis as the purpose of the pretest was to refine the oral administration procedure. Initially, participants were given the option of choosing to receive the oral SUS items as individual voice notes or completing the questionnaire through a WhatsApp call. Based on participant feedback, the voice notes option was removed because listening to multiple recordings before responding was considered tedious. The call format was retained as the sole option as it allowed the oral condition to more closely resemble an interviewer-administered questionnaire where participants could request clarification in real time. Two participants mentioned the word "cumbersome" as confusing or unnatural; this term was known to be a source of difficulty [14]. Participants also mentioned that converting the answer into a 1-5 scale in their minds was "an extra step"; in the final study design, we instructed participants to respond in verbal categories (the options: Strongly Disagree, Disagree, Neutral, Agree, Strongly Agree) rather than numeric values.

## 3.7 Participant Allocation

This study was a part of a larger study evaluating the SUS in a South African context. This larger study ran in parallel with two other conditions (Written Afrikaans SUS and Written isiXhosa SUS). Participant allocation for the larger study was automatically managed through a Microsoft List, which was synchronised with the Microsoft Forms consent and demographic form. Participants were initially separated based on home language(s) and English proficiency. Participants who reported isiXhosa or Afrikaans as their home language or indicated native or fluent proficiency in those respective languages were assigned to the corresponding written SUS condition. Participants who did not meet those specific language criteria were allocated to the Oral English SUS condition. Recruitment was monitored to ensure each language condition met its intended sample size of 25. Once a language condition reached its target, the remaining eligible participants were assigned to the oral condition. For the current study, the analysis included only participants assigned to the Oral English SUS condition. These participants completed both Written English and Oral English SUS conditions, which are the two administration modes examined in this paper.

## 3.8 System Allocation

The six systems were distributed across the written and oral conditions so that the systems varied across conditions. Systems were allocated randomly to the two administration conditions. The system allocation was intended to reduce the influence of any single system being excessively associated with one administration mode. Due to participant non-response during recruitment, the final allocation was imbalanced across systems and administration modes. We address this as a limitation of the study (Section 5.4).

## 3.9 Procedure

Participants were first asked to join the WhatsApp Community via a link or QR code. The Community announcement channel was used to provide information about the study and instructions to complete the demographic and consent form in their preferred

language (English, Afrikaans, or isiXhosa). The English version of the demographic and consent form is provided in Appendix F. The participant information and instruction messages are provided in Appendix E. The demographic and consent information was collected through a single Microsoft Forms document. Participants provided linguistic information, including their English proficiency and home language(s), in the demographic form. For the larger study, participants were automatically allocated to the relevant language condition based on their reported home language(s) and language proficiency. The current study focuses on participants who completed the Written English SUS and Oral English SUS conditions. After completing the demographic and consent form, participants were contacted directly through WhatsApp. The researcher provided an overview of the two study phases and explained the procedure, including the post-session questions that would follow the oral administration. In Phase 1, participants were provided with a single Microsoft Forms link containing the three SUS questionnaires for three allocated systems. The link was prefilled with the allocated systems to avoid participants completing the SUS for the incorrect systems. Once all three written SUS questionnaires had been completed, the researcher contacted participants through WhatsApp to provide instructions for Phase 2 and to arrange a suitable time for the oral session. In Phase 2, participants completed the SUS for the remaining three systems through a WhatsApp call. Participants were encouraged to ask for clarification on any items or ask any questions during the call. Each SUS item was read aloud using the "Standard" English wording while the researcher recorded the participant's response in a separate Microsoft Form document. Participants were then asked post-session questions about their experience with the oral administration and the differences they perceived between the written and oral conditions. (Appendix C)

## 3.10 Research Hypotheses and Variables

The independent variable was the administration mode (written vs. oral). The dependent variable was the SUS score. The following hypotheses were specified for RQ1:

- **H1:** **$H1_0$:** There is no statistically significant difference in mean SUS scores between oral and written administration.
- **H2:** **$H1_a$:** Mean SUS scores differ between oral and written administration.

## 3.11 Quantitative Analysis Pipeline

All quantitative analyses were conducted in R. SUS scores were calculated using Brooke's original SUS scoring formula, with negatively worded items reverse-scored prior to calculating SUS scores. Descriptive statistics were calculated for SUS scores across administration mode and systems. This included the mean, standard deviation, median, interquartile range, minimum, and maximum. The written SUS scores for participants who reported English as a home language were used as a reference for the written "Standard" English SUS for the evaluated systems in the South African context. A paired-samples t-test was conducted at a participant level to test for a difference in mean SUS scores between oral and written conditions. This was done as each participant completed both administration conditions. Due to the small number of participants, the Wilcoxon signed-rank test was used as a non-parametric robustness check. Effect size was assessed using Cohen's $d_z$, together with the confidence interval for the mean difference. A linear mixed-effects model was fitted to account for differences between systems and repeated observations from the same participants. This was done because participants evaluated multiple systems and the allocation of systems across administration conditions was not perfectly balanced. The model included administration mode and system as fixed effects and participant as a random intercept. The written condition was used as a reference for administration mode. The model was used to examine differences between the oral and written SUS scores while accounting for differences between systems and participants. Internal consistency was assessed for both the written and oral conditions using Cronbach's alpha [3, 25]. This

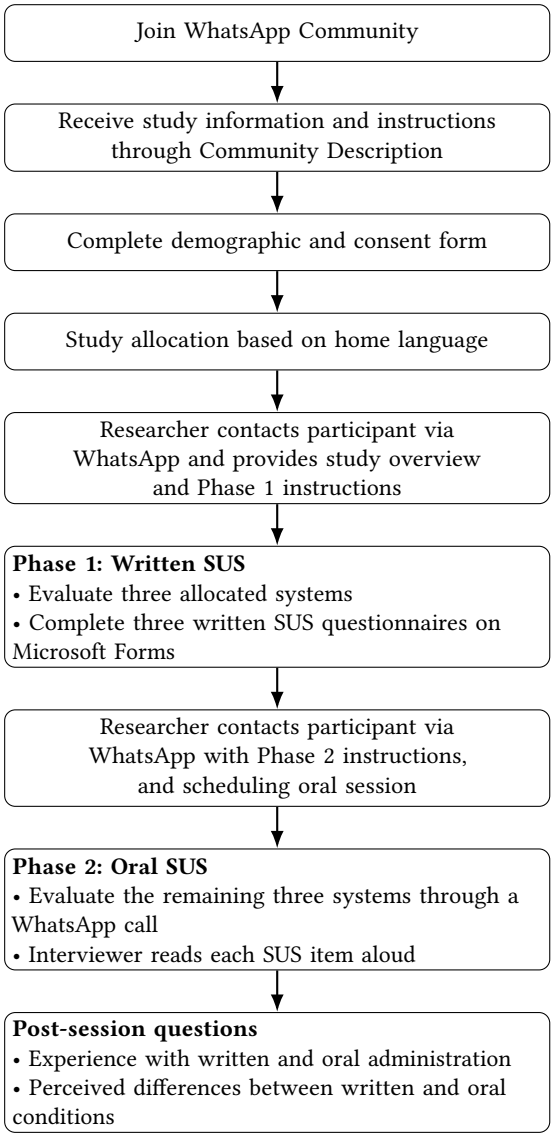



**Figure 1: Participant experience and study procedure**

was used to determine whether the SUS demonstrated acceptable internal consistency across administration conditions. Due to the small number of participants and the sample-dependent nature of internal consistency estimates, the differences in Cronbach's alphas were interpreted descriptively.

### 3.12 Qualitative Analysis (Oral Condition)

Qualitative responses collected during the oral condition's sessions consisted of both the open-ended questions asked at the end of the session and any comments made during administration. These data were analysed using Reflexive Thematic Analysis (RTA) [5], following Braun and Clarke's six-phase framework: Familiarisation with the data, generation of initial codes, searching for themes, reviewing themes, defining and naming themes, and producing the final analysis. Coding was conducted inductively, allowing for themes to emerge from the data rather than from a predetermined framework. The analyst's own positionality and interpretive judgement were treated as part of generating themes rather than a source of bias. The resulting themes were used to help interpret the quantitative findings by providing insight into participants' comprehension, engagement, and experiences of the oral condition.

*3.12.1 Researcher Positionality.* The researcher who conducted the oral interview sessions and qualitative analysis is a monolingual English speaker. While some participants were peers of the researcher, others were not, resulting in varying degrees of familiarity between the researcher and participants. As a monolingual English speaker, the researcher acknowledges that they may have limited insight into the experience of having English as an additional language. This includes the ways in which participants draw from their multilingual repertoires when interpreting and responding to the SUS items. This was considered throughout the qualitative analysis, particularly when interpreting the participants' experiences with both administration conditions and their comprehension of the SUS items. The researcher recognises that participants may experience the administration modes differently and that these differences may reflect participants' individual preferences and the characteristics of each mode.

## 4 Results

### 4.1 Participant Details

The final dataset consisted of 25 participants and 150 system SUS observations. Of these 25 participants, 17 were UCT students, five were students at other universities, two were unemployed, and one was employed. Each participant completed the SUS in both oral and written conditions, with three systems per condition. This resulted in 75 written observations and 75 oral observations. Participants reported three different home languages, with English as the majority (n = 23), while one participant reported isiZulu and one participant reported Shona. Due to the limited representation of non-English home languages, these groups were used descriptively rather than for statistical inference. The distribution of systems evaluated across administration conditions is shown in Table 2. The allocation of individual systems across conditions was not perfectly balanced.

**Table 2: Distribution of systems evaluated across administration conditions**

| System | Oral | Written | Total |
|---|---|---|---|
| ATM | 12 | 16 | 28 |
| EskomSePush | 13 | 11 | 24 |
| Google Search | 15 | 11 | 26 |
| POS Machine | 12 | 10 | 22 |
| Takealot | 13 | 13 | 26 |
| Uber | 10 | 14 | 24 |
| **Total** | **75** | **75** | **150** |

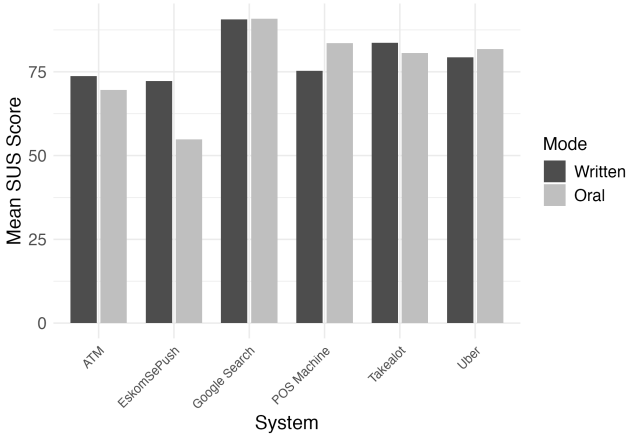



**Figure 2: Mean SUS scores for each evaluated system for oral and written administration**

### 4.2 Mean SUS Scores by System and Administration Mode

**Table 3: Mean and standard deviation of SUS scores by system and administration mode**

| System | Oral | | | Written | | |
|---|---|---|---|---|---|---|
| | *n* | Mean | SD | *n* | Mean | SD |
| ATM | 12 | 69.6 | 23.3 | 16 | 73.8 | 13.8 |
| EskomSePush | 13 | 54.8 | 23.2 | 11 | 72.3 | 16.0 |
| Google Search | 15 | 90.8 | 10.5 | 11 | 90.7 | 8.1 |
| POS Machine | 12 | 83.5 | 16.2 | 10 | 75.2 | 13.3 |
| Takealot | 13 | 80.6 | 7.5 | 13 | 83.7 | 10.7 |
| Uber | 10 | 81.8 | 8.2 | 14 | 79.3 | 14.7 |

The SUS scores varied across systems and administration modes (Table 3; Figure 2). Google Search had the highest mean SUS score in both conditions, while EskomSePush had the lowest mean SUS score in both conditions. The difference between oral and written mean scores differed across systems. Oral scores were lower for ATM, EskomSePush, and Takealot, but were higher for POS Machine and Uber. Google Search showed almost identical mean scores across both conditions.

## 4.3 English Home Language Written Reference

**Table 4: English-home-language written reference SUS scores by system**

| System | n | Mean | SD |
|---|---|---|---|
| ATM | 14 | 75.4 | 11.7 |
| EskomSePush | 10 | 70.0 | 14.9 |
| Google Search | 11 | 90.7 | 8.1 |
| POS Machine | 10 | 75.2 | 13.3 |
| Takealot | 12 | 82.9 | 10.9 |
| Uber | 12 | 82.3 | 12.4 |

The written SUS scores of participants who reported English as a home language were used as a reference for the written "Standard" English SUS. These scores provide a descriptive reference for the SUS scores that may be expected for these systems for participants who reported English as a home language.

## 4.4 Descriptive Statistics

**Table 5: Descriptive statistics for SUS scores by condition.**

| Condition | $n$ | Mean | SD | Median | IQR | Min | Max |
|---|---|---|---|---|---|---|---|
| Oral English | 75 | 77.0 | 19.8 | 80.0 | 20.0 | 22.5 | 100 |
| Written English | 75 | 79.0 | 14.1 | 80.0 | 18.8 | 42.5 | 100 |

The oral condition had a mean SUS score of 77.0, compared to 79.0 for the written condition (Table 5). Oral scores showed greater variability with a larger standard deviation and a wider range than written scores.

## 4.5 Within-Participant Comparison

**Table 6: Within-participant comparison of oral and written SUS scores**

| Measure | Value | $p$-value |
|---|---|---|
| Mean difference (Oral – Written) | -1.93 | |
| SD of differences | 11.34 | |
| 95% CI | [-6.61, 2.75] | |
| Paired-samples $t$-test | -0.853 | .402 |
| Cohen's $d_z$ | -0.17 | |
| Wilcoxon signed-rank test | 103 | .184 |

Since each participant completed both administration conditions, participant-level mean SUS scores were calculated for written and oral conditions, and a paired-samples t-test was conducted to compare scores between conditions. Figure 3 shows the mean SUS score for each participant under both administration conditions, with lines connecting the two scores for each participant. The mean SUS score was 1.93 points lower in the oral condition than in the written condition (Table 6). This difference was not statistically significant ($p$ = .402), with a 95% confidence interval ranging from -6.61 to 2.75.

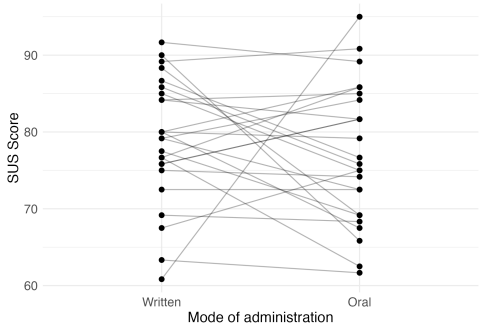



**Figure 3: Participant-level mean SUS scores for oral and written administration conditions**

The Cohen's $d_z$ of -0.17 corresponds to a small within-subject effect. The Wilcoxon signed-rank test produced a similar result and was not statistically significant ($p$ = .184).

## 4.6 Mixed-Effects Model

**Table 7: Fixed effects from the linear mixed-effects model predicting SUS scores**

| Predictor | Estimate | SE | $t$ | $p$ |
|---|---|---|---|---|
| Intercept | 73.026 | 2.955 | 24.711 | $< .001$ |
| Oral administration | -2.493 | 2.404 | -1.037 | .302 |
| EskomSePush | -8.850 | 4.075 | -2.172 | .032 |
| Google Search | 19.181 | 3.996 | 4.800 | $< .001$ |
| POS Machine | 8.116 | 4.175 | 1.944 | .054 |
| Takealot | 10.324 | 3.984 | 2.592 | .011 |
| Uber | 8.326 | 4.066 | 2.048 | .043 |

*Reference levels: Written administration and ATM.*

The model did not show a statistically significant effect of administration mode, with oral administration associated with a 2.49-point lower SUS score than written administration ($F(1, 124.93) = 1.0753$, $p = .302$). There was a statistically significant overall effect of system on SUS scores ($F(5, 127.83) = 10.7838$, $p < .001$).

## 4.7 Internal Consistency

**Table 8: Cronbach's alpha by condition.**

| Condition | Cronbach's $\alpha$ |
|---|---|
| Written English | 0.878 |
| Oral English | 0.931 |

The written condition showed high internal consistency, while the oral condition also showed high internal consistency. These results indicate that the SUS items demonstrated good internal consistency under both conditions. The higher alpha observed for the oral condition should be interpreted descriptively, given that Cronbach's alpha is dependent on the sample and observed item responses.

## 4.8 Qualitative Findings

Reflexive thematic analysis [5] was used to examine participants' comments during oral administration as well as responses to post-session questions about their experience with the oral and written SUS. Four themes were identified:

(1) Administration Affordances
(2) Wording and Comprehension
(3) Interview Influence on Engagement and Response
(4) Perceived Equivalence Across Administration Modes

*4.8.1 Administration Affordances.* Written administration was commonly associated with greater control over the response process, including the ability to reread questions, take additional time to consider or reconsider responses, and use external resources to clarify any confusion (n = 16). P16 noted that the written version provided "less of a time limit" and allowed them to do their own research to clarify any confusion. The oral administration provided immediate opportunities to ask for clarification and receive feedback (n = 12). P20 preferred the oral administration because "the interviewer gets to explain what certain things mean during the questionnaire", which reduced the need to use external sources for clarification. Some participants described the oral condition as quicker or easier to complete (n = 6), while others noted the faster pace reduced the time available to think about their responses (n = 4).

*4.8.2 Wording and Comprehension.* Participants generally found the questionnaire understandable (n = 19), although several participants found specific wording choices that were unfamiliar or difficult to interpret. The term "cumbersome" in item 8 was repeatedly flagged as an unusual choice of wording (n = 10), including by participants who reported having fluent or native English proficiency. P1 described "cumbersome" as "a weird choice to use" and noted that "Native (English) speakers might not even know what that means." Participants also identified issues with particular items, including interpreting questions about imagining how other people would use a system (item 7) (n = 3). P9 described the question as being broad, stating that "Different groups of people will struggle with different applications". Other participants experienced difficulty with questions that contained more technical wording, such as item 5 (n = 2). P19, a participant who reported Shona as a home language, found that the phrase "various functions were well integrated" was confusing. These responses suggest some comprehension difficulties may have been related to the wording of the SUS items rather than just the administration mode.

*4.8.3 Interviewer influence on Engagement and Response.* Participants reported both positive and negative effects of the interviewer on their response experience. Regarding positive effects, participants described the oral condition as more human and conversational, and some felt more comfortable justifying their answers (n = 5). P17 described the interview as making them feel "more confident speaking to a person" and said it "felt human". Similarly, P24 described the spoken version as more interesting, as they could provide their reasoning, and that there was an opportunity to "express thoughts and ideas" about the questions. The interviewer also provided an opportunity to clarify any questions during the process (n = 12). Some participants reported negative effects, such as the presence of the interviewer creating pressure to respond more quickly or feeling like their responses were being judged (n = 4). P1 described the oral administration as creating "more pressure to answer quickly". P2 felt that answering orally created "more pressure by the interviewer".

*4.8.4 Perceived Equivalence Across Administration Modes.* Participants generally perceived the underlying questionnaire as equivalent regardless of the administration mode (n = 21). Several participants recognised that the oral and written conditions asked the same questions. P19 stated that "it feels like the same questions, no difference." Many reported they gave similar responses across conditions (n = 17), with P4 stating "I answered them (administration modes) the exact same way". However, participants reported that responses were not always identical after receiving clarification in the oral administration (n = 4). This suggests that, while participants generally perceived that both administration modes asked the same underlying questions, the interaction with the interviewer may sometimes influence how a participant interprets and responds to an item.

# 5 Discussion

## 5.1 Administration Mode and SUS Scores

In this sample, the administration mode did not result in a statistically significant difference in the SUS scores. The SUS scores differed significantly across systems, indicating variation in SUS scores across the systems evaluated. This is consistent with participants evaluating different systems that had different perceived levels of usability. Both administration conditions demonstrated high internal consistency, indicating that the SUS items were internally consistent under both conditions. These results are consistent with previous research on other psychometric instruments, which found generally similar scores and internal consistency, but also identified potential variation in responses between administration modes [19, 35]. Overall, there was no statistically detectable difference in SUS scores between written and oral administration.

## 5.2 Interviewer Influence and Participants' Experience with Administration Modes

Participants described differences between the written and oral administration modes in how they comprehended and responded to SUS items. These difficulties were mentioned by participants who reported fluent or native English proficiency, which suggests that some challenges with the SUS items may emerge from the wording itself rather than language proficiency alone. Oral administration allowed participants to seek clarification when they encountered these difficulties with specific items. Four participants mentioned that they changed their response to an item after receiving clarification during the oral session. This highlights a potential trade-off with interviewer administration of the SUS: participants can ask for clarification when an item is unclear, but the interviewer may also influence their responses. Previous research has identified interviewer effects in interviewer-administered questionnaires [4, 9, 16]. Other research has also identified that interviewer-administration

has benefits such as allowing for real-time clarification [11]. Together, these findings indicate that administration mode can influence the response process and how the participant experiences the SUS even when overall SUS scores do not differ significantly.

## 5.3 WhatsApp as Survey Medium

WhatsApp was useful as a communication and scheduling medium for this study. The community announcement channel provided a centralised way to communicate with participants without sharing their contact details with other participants in the community. This did have some limitations, such as the announcement channel not allowing old messages to be displayed to new members, which required a researcher to send manual reminders to complete the consent form. The written SUS was administered through Microsoft Forms, as WhatsApp did not provide a direct means of implementing a scale-based questionnaire. The platform did allow scheduling the oral session much more easily, as well as allowing the session to be conducted remotely, although recording the session did require a secondary device because WhatsApp does not allow audio to be recorded through the device's screen-recording functionality. These experiences suggest that WhatsApp can be useful for participant communication and capturing qualitative data remotely, but administering scale-based questionnaires requires an external platform to be used.

## 5.4 Limitations

There were several limitations to the study. The statistical findings should be interpreted cautiously due to our small sample size of 25 participants. English proficiency was self-reported by participants and may therefore not accurately represent their level of proficiency. The use of snowball recruitment also increased the likelihood of recruiting native English speakers. Additionally, the majority of the participants were university students, who may have higher levels of English proficiency and be more familiar with technology and completing forms online than the general public. This limits the generalisability of the findings to participants with different educational and technological backgrounds. The final oral sample had limited representation of fluent or native Afrikaans and isiXhosa speakers. Although these groups were initially planned to be included, recruitment constraints and difficulties scheduling oral sessions meant that the intended language samples could not be achieved within the available time frame. This may have introduced selection bias and limits the extent to which the findings can be interpreted across different linguistic backgrounds. The fixed ordering of Phase 1 and Phase 2, with the written administration always taking place before the oral administration, may have introduced ordering or familiarity effects. This ordering was done to allow participants to encounter any potential misunderstandings with the SUS items in the written condition before having the opportunity to request clarification during the oral condition. Systems were randomly allocated across conditions, but participant non-response resulted in an uneven distribution of systems between conditions (Table 2). Participants also evaluated different systems across the two conditions, meaning that differences in perceived usability between systems may have contributed to differences between conditions. The allocation was intended to reduce the association between certain systems and a specific administration mode, but it does not guarantee balance in a sample this small. The selected systems may have some unknown biases related to participants' individual experience with those systems. The presence of an interviewer during oral administration may have influenced participants' responses. Participants may have felt pressure or altered their responses due to responding directly to an interviewer but also had opportunities to ask for clarity. This is relevant when interpreting the qualitative findings regarding interviewer influence.

## 5.5 Challenges Encountered During the Study

Several challenges were encountered during the study. Ethics clearance affected recruitment and data collection due to receiving delayed approval from the UCT Ethics Committee. This constrained the time available for recruitment and data collection. The sample also had limited variation in English proficiency, as more participants reported having native or fluent English proficiency. This led to a research question about the effects of the interviewer-administered version of the SUS with low-proficiency English speakers being removed from the study. Since participation was voluntary and no compensation was provided, some participants did not complete the study after recruitment. A few participants were reluctant to complete the oral condition due to it requiring a call, and they were unfamiliar with the interviewer. Scheduling a time for the oral session did prove troublesome, as participants required several follow-ups before a suitable time could be arranged due to their own commitments.

## 5.6 Implications and Future Research

The results have some implications for administering the SUS in South African contexts. Table 4 shows SUS scores for the written "Standard" English administration of the SUS among participants who reported English as a home language. These values provide a descriptive reference for the SUS scores the systems evaluated in this study, but due to the small sample, they should not be treated as a baseline. The lack of statistically detectable difference in SUS scores between administration modes, together with the high internal consistency across both conditions, suggests that oral administration may be a useful alternative for the SUS when participants benefit from hearing questionnaire items. This should be interpreted carefully due to the small sample and the fact that most participants reported having fluent or native English proficiency. Oral administration allows for clarification, but it may also introduce interviewer influence. This suggests that administration mode should also be considered in relation to how participants engage with and interpret individual items. Future research should examine these findings using larger, more linguistically diverse samples. Research using other psychometric instruments demonstrated the feasibility of interviewer-administered measures among participants with low literacy [34]. This supports further investigation of the oral SUS administration in populations where there is a greater variation in literacy levels. Future studies could investigate whether simplified SUS wording, such as the version developed by Holden [15] for the cognitively impaired, or translated versions could improve accessibility of the SUS in South African contexts.

## 6 Conclusions

This study examined whether the oral administration of the "Standard" English SUS affected SUS scores and internal consistency compared to the written administration. It also explored participants' experiences with the interviewer-administered SUS. The quantitative findings showed no statistically significant difference in SUS scores between modes, and both conditions demonstrated high internal consistency. Within this sample, the oral administration produced similar SUS scores to those obtained through the written administration. Participants experienced the two modes differently. The written administration was described as having greater control over the response process, while the oral administration allowed for clarification and was described as more conversational and engaging. Interviewer presence could also introduce pressure or influence responses. Thus, the administration mode may affect how participants interpret and respond to the SUS items, even if there was no statistically detectable difference in overall SUS scores. Overall, oral administration may provide a useful alternative to the written "Standard" English administration of the SUS. These findings should be interpreted cautiously given the small sample, which consisted mostly of participants who reported fluent or native English proficiency. Further research with larger and more linguistically diverse samples is needed to determine whether similar results occur among participants with different levels of English proficiency and literacy. Further research could also examine the simplified version of the SUS in multilingual contexts such as South Africa.

## 7 AI Usage Statement

Generative AI was used to provide feedback on the structure of the paper and to provide criticism on my arguments. It was used to help code the tables for the quantitative analysis in LaTeX as well as the code for the visualisations done in R. The writing was done solely by me.

## 8 Acknowledgement

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# A "Standard" English System Usability Scale

The following is the "Standard" English version of the System Usability Scale (SUS) used in both administration modes in the study.

(1) I think that I would like to use this system frequently.
(2) I found the system unnecessarily complex.
(3) I thought the system was easy to use.
(4) I think that I would need the support of a technical person to be able to use this system.
(5) I found the various functions in this system were well integrated.
(6) I thought there was too much inconsistency in this system.
(7) I would imagine that most people would learn to use this system very quickly.
(8) I found the system very cumbersome to use.
(9) I felt very confident using the system.
(10) I needed to learn a lot of things before I could get going with this system.

Each item is rated on a five-point Likert scale ranging from 1 (Strongly disagree) to 5 (Strongly agree).

# B System Task Instructions

**Table 9: Tasks completed by participants for each system evaluated.**

| System | Task |
|---|---|
| Uber | Get to the fare estimate screen for a trip of your choice. You do not need to complete the booking. |
| Takealot | Search for a product you would consider buying, and either add it to your cart or your wishlist. |
| EskomSePush | Add your area, then check the current load-shedding schedule for it. |
| ATM | Next time you withdraw cash, pay attention to the experience. |
| Google Search | Search for anything you are curious about. |
| POS Machine | Next time you pay by tapping or inserting your card, pay attention to the experience. |

# C Post-Session Questions

(1) Can you describe in your own words what this questionnaire is trying to measure?
(2) Did any words or phrases feel unnatural, awkward, or confusing in either version?
(3) Were there any questions where you weren't sure what was being asked?
(4) Did the written and spoken versions ask the same thing, or did anything feel different between them?
(5) Which version felt easier to understand or answer, and why?
(6) Did you answer any questions/items differently between the two? What caused that?
(7) Which version would you prefer in a real system evaluation questionnaire, and why?

# D Recruitment Material

The following recruitment materials were used to recruit participants for the larger study consisting of all the oral SUS, written Afrikaans, and written isiXhosa conditions.

## D.1 Recruitment Message

Hi there! Are you a speaker of English, isiXhosa, or Afrikaans?

We're running a UCT research study where participants evaluate the usability of a few different apps or devices in their own language, as well as a second language (for multilingual participants) in order to study the effects that language may have on the results.

We're looking for people who: - Are 18 years or older - Fall into one of the following groups: 1. English speakers at any level of proficiency (from basic to fluent speakers) 2. isiXhosa native speakers (first-language or home-language) 3. Afrikaans native speakers (first-language or home-language)

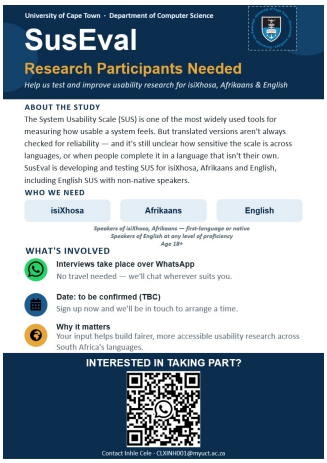
What you will do: - Evaluate 6 different apps or devices using a short survey - Participate in a short interview afterwards about your experience The entire process should take about 15-20 minutes, split into two 7-minute surveys and a short 5-minute interview afterwards. Everything will happen online via WhatsApp and Microsoft Forms, so you can complete the study in your own time.

If you are interested, you can join our WhatsApp community here: [Study Community Link]

Once you join, please complete the consent form found in the community description. A researcher will contact you with further details soon after!

Feel free to email me if you have any questions. (Researcher's Email Address)

## D.2 Recruitment Poster




*Recruitment poster used for participant recruitment.*

# E Participant Information and Instruction Messages

## E.1 Study Information

Welcome! Thanks for joining the SUSEVAL usability study.

This is an Honours research project conducted from the Computer Science Department of UCT.

Step 1:

Complete the short consent + demographic form:

- English: [English Consent Form Link]
- isiXhosa: [isiXhosa Consent Form Link]
- Afrikaans: [Afrikaans Consent Form Link]

Step 2:

A group admin will DM you directly with your personalised instructions; you won't need to do anything else in this group chat.

Privacy:

Your number stays private from other participants. All study communication happens one-on-one.

Any questions? Just message a group admin directly.

## E.2 Phase 1: Written SUS Instructions

**Thanks for joining the SUSEVAL study! Here's how it will work.**

You'll be evaluating **6 everyday apps/tasks** in two stages. There are no right or wrong answers. We're interested in your honest experience.

**Phase 1: Written English (3 tasks)** For each of these, I'll describe 3 short tasks, then ask you to fill in a 10-question form about the experience for each task. It should take about 5 minutes each. If a task involves a physical thing (like an ATM or Point of Sale (POS) Machine), just try it next time you naturally would, then fill in the form afterwards. No need to do it on the spot if you don't remember how your experience was with that system.

These are your 3 systems/tasks to evaluate:

(1) System 1: Task 1
(2) System 2: Task 2
(3) System 3: Task 3

Link to Questionnaire: [Link to Prefilled Microsoft Forms document]

**Phase 2: Oral English (3 tasks)**

Once I've received your Phase 1 responses, I'll follow up shortly after with Phase 2. It is the same idea, but this time I'll ask the questions by a short phone call instead of a form, and you're welcome to ask me to clarify anything along the way.

**A few follow-up questions**

Once you've finished both phases, I may reach out to ask if you'd like to share a bit more about your experience- a short voice note, text response, or call, just 5-10 minutes, about what felt easy or confusing.

Everything you share stays confidential and is only used for this research. You can stop at any point with no penalty. Let me know if anything's unclear before we start!

## E.3 Phase 2: Oral SUS Instructions

Thanks for completing Phase 1! Here's Phase 2.

Same idea as before, but this time for these 3 systems/tasks:

(1) System 4: Task 4
(2) System 5: Task 5
(3) System 6: Task 6

We can arrange a **short phone call** at a time suitable for you to complete the questionnaires for these systems. You can ask for clarification on any item, any time. Afterwards, I will ask a few follow-up questions about your experience.

Same rating scale as before: **1 = Strongly Disagree, 2 = Disagree, 3 = Neutral, 4 = Agree, 5 = Strongly Agree.**

Let me know when it is suitable to call you, and we'll get started whenever suits you.

# F Consent and Demographic Form

The blank consent and demographic form used for participants is shown below. This is directly saved from Microsoft Forms.

# Demographic & Consent form English

These questions help us understand who took part in this study and whether language background affects how people experience the SUS questionnaire. Responses are confidential and used for research purposes only — you may skip any question you'd prefer not to answer.

* Required

## Consent form

**Invitation to participate, and benefits:** You are invited to participate in a research study conducted with users of digital systems. The study aims to evaluate the suitability of the System Usability Scale in South Africa. I believe that your experience would be a valuable source of information, and hope that by participating you may gain useful knowledge.
**Procedures:** During this study, you will be asked to complete six surveys in which you rate your experiences using a system.
**Recording (Oral phase only):** If you are selected to participate in the Oral phase, you will be asked to respond via a short WhatsApp call. If you consent to audio recordings this audio will be used only to extract your survey responses for analysis. Audio recordings will not be stored after transcription and will not be shared or released publicly. If you do not consent to audio recordings we will just transcribe during the call.
**Risks:** This study uses third-party platforms. WhatsApp for communication and Microsoft Forms for written surveys. WhatsApp Communities do not expose participants' phone numbers to other members, and Microsoft Forms does not require sign-in or link responses to your identity. However, as with any third-party platform, there is a minor residual privacy risk inherent in data transmitted over these services, outside the researcher's direct control.
**Feedback**: You will receive feedback about the results of this research via the same WhatsApp groups in which you will receive the links to complete the surveys.
**Disclaimer/Withdrawal:** Your participation is completely voluntary; you may refuse to participate or withdraw at any time before your data has been anonymised, without having to state a reason and without prejudice or penalty. Once your data has been anonymised, it can no longer be linked back to you, and so it will not be possible to remove it individually. Should you withdraw before this point, the researcher commits not to use any of the information you have provided. Note that the researcher may also withdraw you from the study at any time.
**Confidentiality**: All information collected in this study will be kept private in that you will not be identified by name or by affiliation to an institution. Confidentiality and anonymity will be maintained as pseudonyms will be used.
**What signing this form means:** By signing this consent form, you agree to participate in this research study. The aim, procedures to be used, as well as the potential risks and benefits of your participation have been explained verbally to you in detail, using this form. Refusal to participate in or withdrawal from this study at any time will have no effect on you in any way. You are free to contact a researcher to ask questions or request further information at any time during this research.

1. Please enter your name *

2. Please enter your phone number *

3. I agree to participate in this study *

- ◯ Yes
- ◯ No

4. I consent to WhatsApp audio recording *

- ◯ Yes
- ◯ No

5. Please enter your name (Digital Signature) *

6. Please enter today's date *

## Section A: Personal details

7. What is your age?

- ◯ 18-24
- ◯ 25-34
- ◯ 35-44
- ◯ 45-54
- ◯ 55+

8. What is your gender?

- ◯ Woman
- ◯ Man
- ◯ Non-binary
- ◯ Prefer not to say
- ◯ Other

9. What is your current occupation/status?

- ☐ UCT Student
- ☐ UCT Staff
- ☐ Student (Other institution)
- ☐ Employed
- ☐ Unemployed
- ☐ Other

10. What is your highest completed level of education?

- ◯ High School
- ◯ Undergraduate degree/diploma
- ◯ Postgraduate degree
- ◯ Other

11. Which province do you currently live in?

- ◯ Eastern Cape
- ◯ Free State
- ◯ Guateng
- ◯ KwaZulu-Natal
- ◯ Limpopo
- ◯ Mpumalanga
- ◯ North West
- ◯ Northern Cape
- ◯ Western Cape

## Section B: Language Background

12. What is your home language - the language you learned first, growing up? (You may select multiple options)

- [ ] English
- [ ] Afrikaans
- [ ] isiXhosa
- [ ] Other

13. Which other languages do you speak, read, or write, even if not fluently?

14. How would you rate your proficiency in isiXhosa

- ◯ None
- ◯ Basic
- ◯ Conversational
- ◯ Fluent
- ◯ Native

15. How would you rate your proficiency in Afrikaans

- ◯ None
- ◯ Basic
- ◯ Conversational
- ◯ Fluent
- ◯ Native

16. How would you rate your proficiency in English

- ◯ None
- ◯ Basic
- ◯ Conversational
- ◯ Fluent
- ◯ Native

17. If English is not your first language, when did you start learning it?

- ◯ Before starting school
- ◯ In primary school
- ◯ In high school
- ◯ As an adult
- ◯ N/A - First language

18. Which language would you be most comfortable completing this study in?

- ◯ English
- ◯ Afrikaans
- ◯ isiXhosa

## Technology Use

19. Do you have regular access to a smartphone with WhatsApp?

- ◯ Yes
- ◯ No

20. How often do you use WhatsApp?

- ◯ Several times a day
- ◯ Daily
- ◯ A few times a week
- ◯ Rarely
- ◯ Never

21. How would you describe your overall comfort using apps or websites?

- ◯ Very comfortable
- ◯ Comfortable
- ◯ Neutral
- ◯ Uncomfortable
- ◯ Very uncomfortable

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